There are two types of pipeline structure :
1) Arithmetic pipeline .
2) Instruction pipeline

pipeline Conflicts :—
There are 5 pipeline Conflict which Cause the pipeline to deviate it's normal performance are :
1) Timing Variation :—
2) Data Hazards
3) Branching
4) Interrupt
5) Data dependancy .

Consider a pipeline having 4 phases with durations 60, 50, 90 and 80 Nano Seconds. The latch delay is 10 NS Calculate pipeline cycle time, Non pipeline execution time, Speed of ratio, pipeline time for 1000 task, Non pipeline time for 1000 task and throughput ?

pipeline cycle time = Max { } + { delay }
Non pipeline execution time for 1 instruction = Σ {all the value}
Speed of ratio = $\frac{NPET}{PCT}$

pipeline time for 1000 task = Time taken for 1st task + time taking for remaining 999 task
=> no. of clock cycle + 999 x 1 cycle.
Non pipeline for 1000 task = 1000 x Time taken for 1 task
throughput = $\frac{\text{Total no. of task}}{\text{pipeline time for 1000 task}}$